
Factors Affecting Bird Extinction

A project investigating the impact of biologic factors on extinction times of birds

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1 Introduction - Hugo

The purpose of this report is to present the results of our statistical investigation of a data set on factors affecting extinction among bird species. Specifically, the report will examine the impact of nesting pairs, size, and migratory status on the time to extinction of bird species. Extinction is a critical issue that has attracted the attention of many researchers and conservationists in recent years. Birds are a vulnerable group of species facing the threat of extinction. The factors affecting extinction are complex and multifaceted, requiring in-depth analysis to develop effective conservation strategies.

While the report aims to provide insight into the factors affecting bird extinction, it is essential to acknowledge the limitations of data. The data set used in this report is specific to breeding pairs of land-bird species collected from 16 islands around Britain over several decades. Therefore, the findings may not be generalizable to other bird species or regions. Additionally, the report will not include a literature background section, focusing instead on statistical analysis and findings.

2 Analysis

2.1 Fitting the model - Snow

The program R Studio and programming language R was used to assume and test linear relationship with the data-set. We started with fitting a model to the provided data-set. There are four variables in the data set, i.e., the average time of extinction (Time), average number of nesting pairs (Pairs), species size (Size) and migratory status (Status). We tried various transformation to find the best fitting model. Our first full model, which was the basis of future transformed model is given below.

$$\hat{Y} = \hat{\alpha} + \hat{\beta}_1 \cdot \text{Pairs} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \hat{\beta}_4 \cdot \text{Pairs} \times \text{Size} + \hat{\beta}_5 \cdot \text{Pairs} \times \text{Status} \\ + \hat{\beta}_6 \cdot \text{Size} \times \text{Status} + \hat{\beta}_7 \cdot \text{Pairs} \times \text{Size} \times \text{Status}.$$

This model has a low p -value which is 0.001446. The smaller the p -value, the stronger the evidence against the null hypothesis, stating that all slope estimates could possibly be equal to 0. Therefore a p -value of less than 1% is very significant. Although we have a very good p -value, it can not tell us how much the variables affect the extinction of bird species. R^2 -value of 0.2542 (see Table 1) does not make it a good fitted model. This means the relationship between the variables only explains 25% of the variation in the data. Furthermore, it was clear from the scatter-plot in Figure 1 that there was no linear relationship. There are many outliers which heavily affect the R^2 -value.

	Estimate	p-value
Intercept	0.7775	0.908
Pairs	1.2397	0.475
Size	0.1759	0.983
Status	4.0489	0.609
Pairs×Size	-0.9058	0.710
Pairs×Status	0.5317	0.792
Size×Status	-9.4013	0.345
Pairs×Size×Status	2.0047	0.465
Residual standard error	9.237	-
Adjusted R^2	0.2542	-
F-statistic	3.97	0.001446

Table 1: Model properties of the full, initial model

2.2 Checking for outliers and influential observations - Sebastian

Now it is time to take a closer look at the full fitted model. First it will be investigated whether or not some observations are affecting the model in such way that the model becomes misleading when trying to explain the outcome variable. Those observations are called influential observations, as they have an influential impact on the model by affecting the estimates of the slope coefficients. To get a visually indication of possible influential observations, a spread-level plot is being presented (see Figure 1). The spread-level plot shows the predicted values of the outcome variable against the standardised residuals stdres_i , which are defined as a standardised version of the residuals $e_i = Y_i - \hat{Y}_i$:

$$\text{stdres}_i = \frac{e_i}{s_e \sqrt{1 - h_{ii}}}. \quad (1)$$

The residual standard error s_e is a measure of the average error made by the regression prediction of the outcome variable. The value h_{ii} is called the leverage of the i th observation. The range of the leverage is defined as $h_{ii} \in [\frac{1}{n}, 1]$, where n is the number of observations in the data set. If the i th prediction of the outcome variable is far away from the sample mean of the outcome, then h_{ii} is going to be larger than if the prediction was closer to the sample mean.

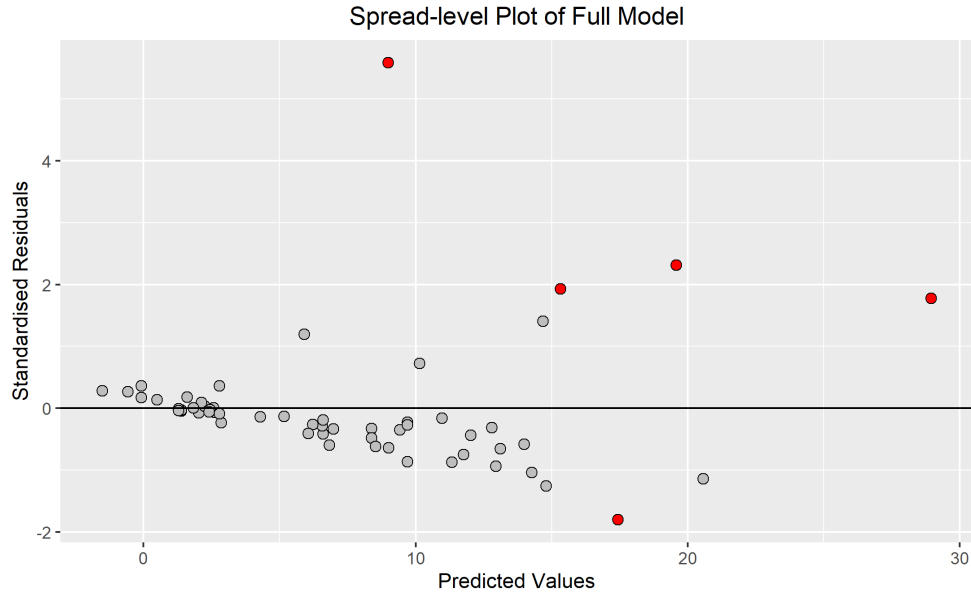


Figure 1: Spread-level plot of the predicted values against the standardised residuals of the full model

In the spread-level plot above (Figure 1) the predicted values with an absolute standardised residual higher than 1.7 has been highlighted in red. The observations corresponding to these predicted values might be considered as outliers, as their standardised residuals are larger than for any of the other predicted values. The highlighted observations are:

i	Species	stdres_i
28	Raven	5.5849
20	Rock Dove	2.3133
29	Skylark	1.9315
26	Jackdaw	1.7967
60	Starling	1.7816

Table 2: Top 5 largest standardised residuals (absolute value)

It can be difficult to identify the outliers directly from the spread-level plot, as it is unclear how large the absolute value of the standardised residual should be to consider an observation as an outlier. Additionally an outlier is not necessarily an influential observation, which actually affects the regression model. Because of these insecurities and to support the visual representation of the outliers in terms of the spread-level plot, the Cook's distance for all observations is being computed. Cook's distance of observation i is defined as

$$D_i = \frac{\text{stdres}_i^2}{(p+1)} \times \frac{h_{ii}}{(1-h_{ii})}, \quad (2)$$

where p is the number of predictors in the model. The leverage h_{ii} and standardised residual stdres_i are well-known measurements from earlier. If an observation has a high value for Cook's distance,

it also tends to be more influential. But how large does the Cook's distance have to be to define an observation as being influential? As for the standardised residuals, this limit is not well-defined. Instead the observations with the highest Cook's distance are being inspected:

<i>i</i>	Species	Cook's dist.
20	Rock Dove	0.3332
28	Raven	0.28
60	Starling	0.2678
26	Jackdaw	0.106
29	Skylark	0.051

Table 3: Top 5 largest Cook's distances

It is noticeable, that the 5 observations with the highest absolute standardised residuals also are the ones with the largest Cook's distance. This relationship is also indicated by (2), where the standardised residual of observation i is in the nominator and squared, which means that the greater standardised residual (absolute value) an observation has, the greater its Cook's distance is.

To decide which observations are actually influencing the model, an observation is removed from the original data set one by one. The first observation removed is the one with the largest Cook's distance and then the next observation according to Table 3 and so on. During this process it is checked whether the properties of the model such as the slope coefficients, adjusted R^2 - and p -value change. When the process of removing an observation does not change the properties of the model, it is decided that this observation and all observations with a smaller Cook's distance, will not influence the model and are not considered as influential observations.

The result of this process was that observations with index $i = 20$ and $i = 28$ were influential, why they were removed from the original data set and the model were updated according to the reduced data set. When trying to remove additional observations, most of the slope estimates remained the same, why the decision to only remove observation number 20 and 28 was made. The slope estimates, residual standard error, adjusted R^2 -value and F -statistic with corresponding p -value of the model are now:

	Estimate	<i>p</i> -value
Intercept	0.7775	0.844
Pairs	1.2397	0.228
Size	0.1759	0.972
Status	2.5904	0.591
Pairs×Size	-0.9058	0.528
Pairs×Status	-0.1536	0.902
Size×Status	-7.9428	0.187
Pairs×Size×Status	2.69	0.11
Residual standard error	5.445	-
Adjusted R^2	0.4654	-
<i>F</i> -statistic	8.339	$8.557 \cdot 10^{-7}$

Table 4: Model properties after removing influential observations

From the bottom of Table 4 it can be seen that the *p*-value of the *F*-test of the model is well below the significance level of 0.05, which means that we can reject the null hypothesis stating that none of the predictors are related to the outcome variable. However, the *p*-values of several slope estimates are well above 0.05. This fact implies that it is not possible to reject the null hypothesis that β_i for $i = 1, \dots, 7$ are equal to 0. This might seem as a contradiction to the low *p*-value of the *F*-test, but in the *F*-test just a single of the slope estimates has to be different than 0, where each *t*-test is performed individually for each slope estimate. So it is fairly likely for each slope estimate to be equal to 0, but it is unlikely that this will be the case for all 7 of them.

2.3 Transformations of the outcome variable - Hugo

Before proceeding with the multiple linear regression, it is essential to ensure that the data satisfies the assumptions of the model. One critical assumption is that the outcome variable (time to extinction) follows a normal distribution. By looking at Figure 2 and at the spread-level plot and qq-plot of the model, before any transformation. It can be observed that the model does not follow a normal distribution. In order to improve the fit of the model, the following three types of transformation are performed on the outcome variable - Time:

- 1) Square-root transformation - $\sqrt{Y} = \alpha + \beta X$
- 2) Logarithmic transformation - $\log(Y) = \alpha + \beta X$
- 3) Inverse transformation - $1/Y = \alpha + \beta X$

The spread-level plots (left side) and qq-plots (right side) of the transformations can be seen on here:

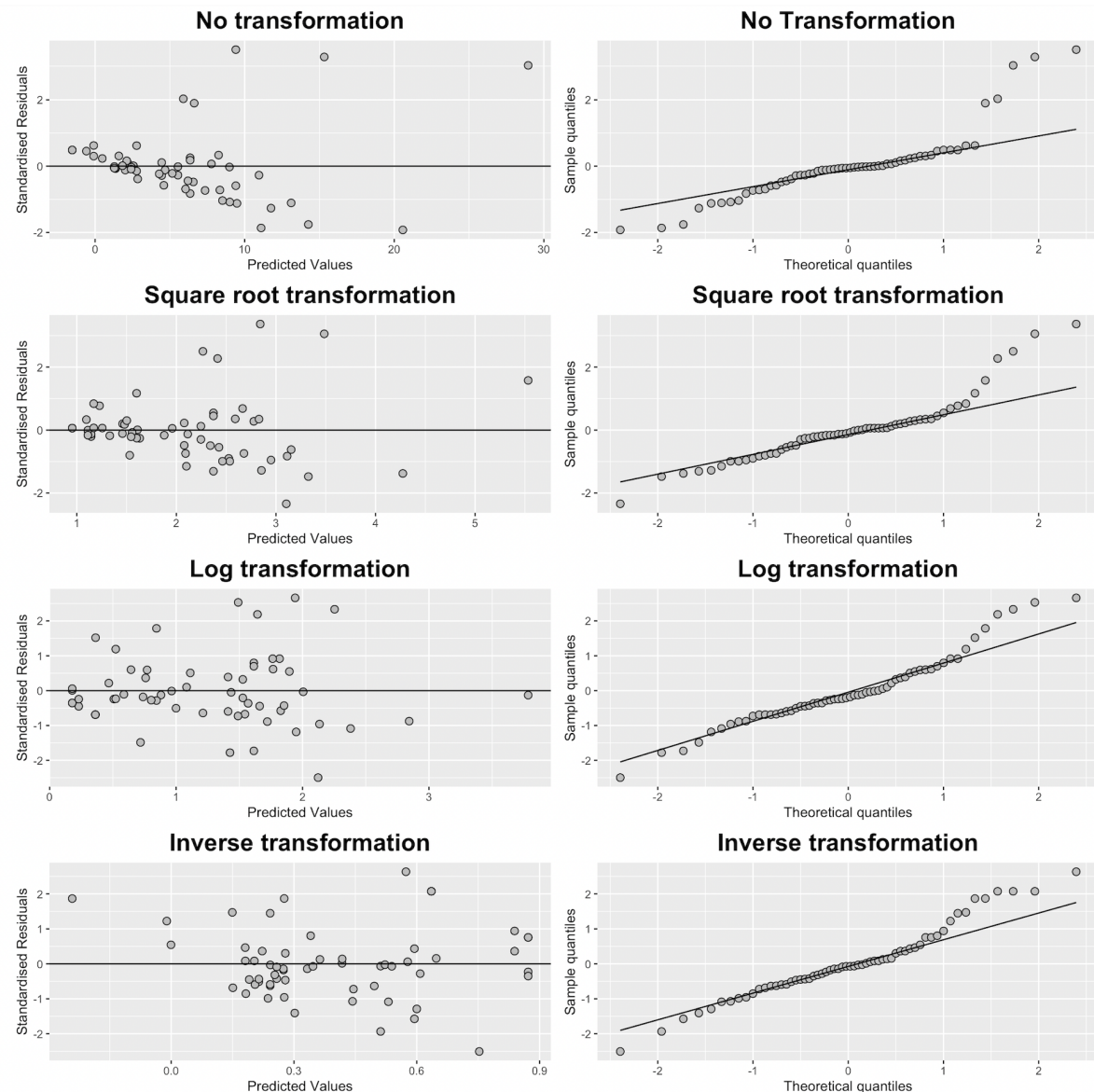


Figure 2: Spread-level plots and qq-plots of the model after each transformation of the outcome variable - Time

Based on visual inspection of all the spread-level plots Figure 2, all of the transformations seem to show that the standardised residuals have about the same variation at every level of the predicted value and they do not follow any patterns - meaning they meet the assumption of homoscedasticity and linearity. However, by examining the qq-plots the logarithmic and inverse transformations appear to provide a better fit than the square root transformation.

To determine which transformation to use, the adjusted R-squared values are evaluated for each model in the model statistics. A summary can be seen here:

Type of transformation	Adjusted R^2	p -value	F -statistic
None	0.4654	$8.557 \cdot 10^{-7}$	8.339
Square root	0.5535	$1.092 \cdot 10^{-8}$	11.45
Log	0.6301	$1.05 \cdot 10^{-10}$	15.36
Inverse	0.6175	$2.406 \cdot 10^{-10}$	14.61

Table 5: Model statistics for different transformations of time

By comparing adjusted R^2 , p -values and F -statistic for both the logarithmic and inverse transformations. For every parameter it can be determined that the logarithmic transformation provides the best fit. The model after the transformation looks as follows:

$$\begin{aligned} \log(\text{Time}) = & \hat{\alpha} + \hat{\beta}_1 \cdot \text{Pairs} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \hat{\beta}_4 \cdot \text{Pairs} \times \text{Size} + \hat{\beta}_5 \cdot \text{Paris} \times \text{Status} \\ & + \hat{\beta}_6 \cdot \text{Size} \times \text{Status} + \hat{\beta}_7 \cdot \text{Pairs} \times \text{Size} \times \text{Status} + \varepsilon_i. \end{aligned}$$

2.4 Transformations of the predictor variables - Anna

To further improve the fit of the model, several different transformations were performed on the predictor variables. Doing so helped to ensure that the assumptions of the regression model are being met, thus making it possible to draw meaningful conclusions from the produced model.

As the predictor variable "Pairs" is the only numerical variable in the model, the transformations were performed on that variable. Doing so on the other variables, "Size" and "Migratory status", would not bring any benefit to the model, as those two predictor variables are categorical.

From Figure 2 in the preceding section it is visible that the data still has a slight positive skew. To combat that, the following transformations were tested on the "Pairs" variable:

- 1) the logarithmic transformation, $\log(\text{Pairs})$
- 2) the inverse transformation $1/\text{Pairs}$
- 3) the square-root transformation $\sqrt{\text{Pairs}}$

The decision on which transformation, if any, is beneficial to the development of the regression model is based on the analysis of the spread-level plot and the qq-plot of the residuals for each of the transformations. Those can be seen in Figure 3.

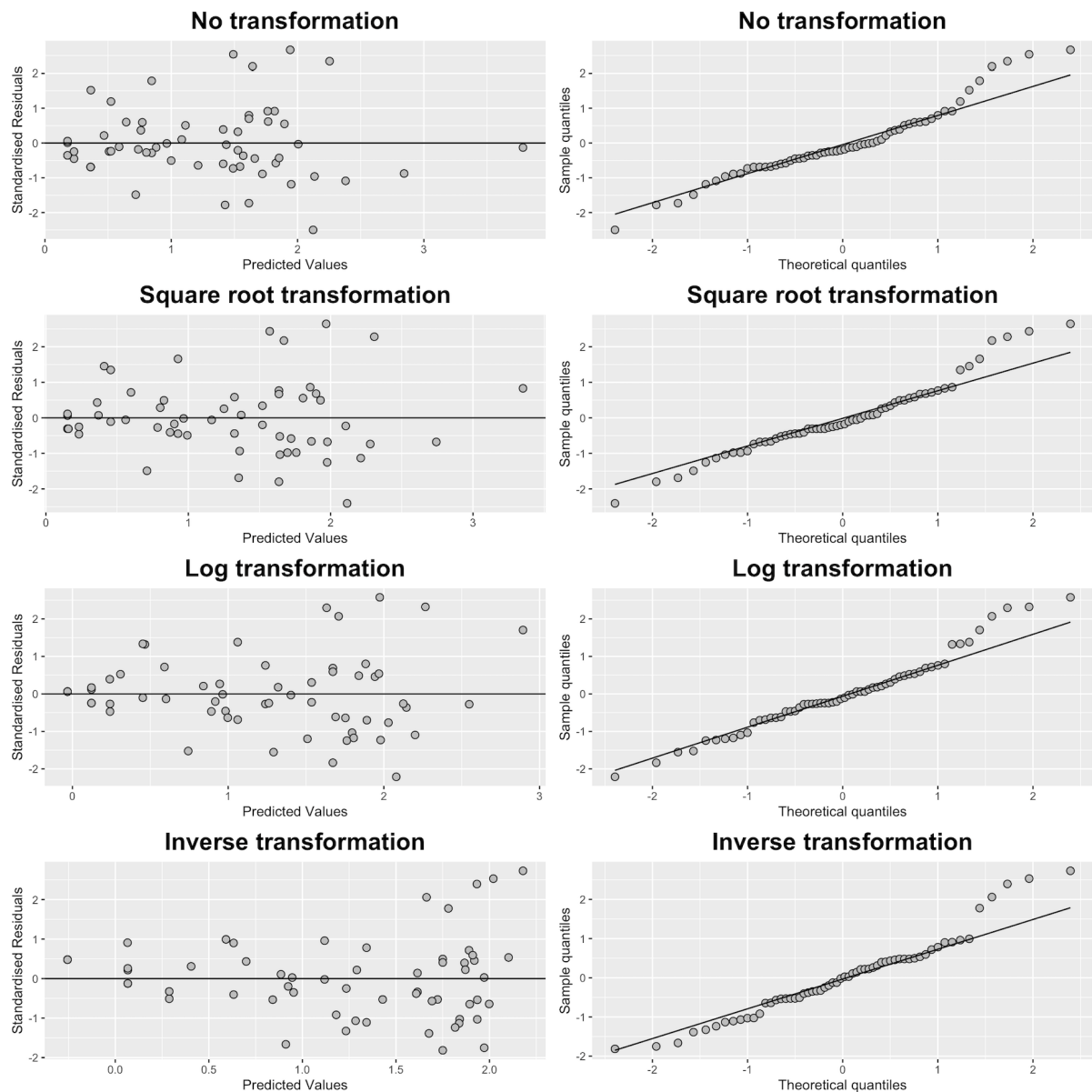


Figure 3: Spread-level plots and qq-plots of the model after each transformation of the "Pairs" variable

Looking at the plots, we can see a slight difference in the spread of the residuals and a change in how close they follow the normal distribution. Those changes are quite minor, the plots look very similar, and for an untrained eye, it is difficult to say if there is any benefit to transforming the predictor variable. Therefore, after each transformation, the summary of the model was recalled, and investigated against the model summary with no transformations on the predictor variables.

Type of transformation	Adjusted R^2	p -value	F -statistic
None	0.6301	$1.05 \cdot 10^{-10}$	15.36
Log	0.6317	$9.427 \cdot 10^{-11}$	15.46
Inverse	0.5596	$7.826 \cdot 10^{-9}$	11.71
Square-root	0.643	$4.354 \cdot 10^{-11}$	16.18

Table 6: Model statistics for different transformations of pairs

When comparing model properties of different transformations (see Table 6) it is more clear, that the square-root transformation is, to a minor degree, beneficial to the accuracy of the model. It is providing a better value for the adjusted R^2 and a larger value for F -statistic. Taking that into consideration, and wanting to work with the predictor transformation for educational purposes, the decision was made to include the square root transformation in the regression model. The model after the transformations looks as follows:

$$\begin{aligned} \log(\text{Time}) = & \hat{\alpha} + \hat{\beta}_1 \cdot \sqrt{\text{Pairs}} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \hat{\beta}_4 \cdot \sqrt{\text{Pairs}} \times \text{Size} + \hat{\beta}_5 \cdot \sqrt{\text{Pairs}} \times \text{Status} \\ & + \hat{\beta}_6 \cdot \text{Size} \times \text{Status} + \hat{\beta}_7 \cdot \sqrt{\text{Pairs}} \times \text{Size} \times \text{Status} + \varepsilon_i. \end{aligned}$$

2.5 Model comparison - Sebastian

After having considered various transformations of both the outcome variable and one of the predictor variables it is time to compare the model with a number of nested models. A model is nested, if it can be obtained from a more complex model involving one or more predictor variables. More precise it possible to obtain the nested model by choosing $\beta_i = 0$ for $i = 1, \dots, p$, where p is the number of predictor variables. Possibly $\beta_i = 0$ for more than one value of i . For the model in this paper the following models will be considered:

Transformed Model

$$\begin{aligned} \log(\text{Time}) = & \hat{\alpha} + \hat{\beta}_1 \cdot \sqrt{\text{Pairs}} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \hat{\beta}_4 \cdot \sqrt{\text{Pairs}} \times \text{Size} + \hat{\beta}_5 \cdot \sqrt{\text{Pairs}} \times \text{Status} \\ & + \hat{\beta}_6 \cdot \text{Size} \times \text{Status} + \hat{\beta}_7 \cdot \sqrt{\text{Pairs}} \times \text{Size} \times \text{Status} + \varepsilon_i. \end{aligned}$$

Basic Model

$$\log(\text{Time}) = \hat{\alpha} + \hat{\beta}_1 \cdot \sqrt{\text{Pairs}} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \varepsilon_i.$$

Model 1

$$\log(\text{Time}) = \hat{\alpha} + \hat{\beta}_1 \cdot \sqrt{\text{Pairs}} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \hat{\beta}_4 \cdot \sqrt{\text{Pairs}} \times \text{Size} + \varepsilon_i.$$

Model 2

$$\log(\text{Time}) = \hat{\alpha} + \hat{\beta}_1 \cdot \sqrt{\text{Pairs}} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \hat{\beta}_5 \cdot \sqrt{\text{Pairs}} \times \text{Status} + \varepsilon_i.$$

Model 3

$$\log(\text{Time}) = \hat{\alpha} + \hat{\beta}_1 \cdot \sqrt{\text{Pairs}} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \cdot \text{Status} + \hat{\beta}_6 \cdot \text{Size} \times \text{Status} + \varepsilon_i.$$

Model 4

$$\log(\text{Time}) = \hat{\alpha} + \hat{\beta}_1 \cdot \sqrt{\text{Pairs}} + \hat{\beta}_2 \cdot \text{Size} + \hat{\beta}_3 \times \text{Status} + \hat{\beta}_7 \times \sqrt{\text{Pairs}} \times \text{Size} \times \text{Status} + \varepsilon_i.$$

As all the models can be obtained from the Transformed Model, they can be considered as nested models of the Transformed Model. Additionally the basic model can be obtained from each of Model 1, Model 2, Model 3, and Model 4, why the basic model is a nested model of all the other models.

To compare the models a F -test is being carried out, which requires the models to be nested. The null-hypothesis of the F -test is that the slope estimate of all the additional predictor variables in the larger model are equal to 0. In other words, that they do not influence the outcome variable by not improving (or weakening) the prediction of the outcome. In the test the F -statistic is being computed, which is defined by:

$$F = \frac{(SS_{Res,S} - SS_{Res,L}) / (df_{Res,S} - df_{Res,L})}{MS_{Res,L}},$$

where SS is the residual sum of squares, df is the degrees of freedom and MS is the mean square. The double subscripts S and L denotes the smaller and the larger model.

If the smaller model is true, it can be assumed that the F -statistic follows a Fisher distribution with $df_1 = p$ and $df_2 = n - p - 1$, where n is the numbers of observations and p is the number of predictors. If the p -value of the F -test is below the significance level of 0.05 the null-hypothesis will be rejected, which means that at least one of the slope estimates of the additional predictors are not equal to 0 with statistical significance. However if the p -value is exceeding the significance level of 0.05, then it is not possible to reject the null-hypothesis and it becomes statistical significant to conclude that all the additional predictors are redundant, i.e $\hat{\beta}_i = 0$ for the relevant values of i .

The F -test is performed on a triplet of models, all nested to each other. The organization of the test appears in Table 7.

# <i>F</i> -test	Models involved	<i>F</i> -statistic	<i>p</i> -value
1.	Basic Model		
	Model 1	3.4267	0.0698
	Transformed Model	1.9003	0.141
2.	Basic Model		
	Model 2	0.3646	0.5486
	Transformed Model	2.921	0.0425
3.	Basic Model		
	Model 3	0.0016	0.9681
	Transformed Model	3.042	0.037
4.	Basic Model		
	Model 4	1.3188	0.2782
	Transformed Model	5.1711	0.0271
5.	Basic Model		
	Transformed Model	2.2819	0.0729

Table 7: Results of *F*-test of nested models

In each of the *F*-tests the Basic Model is compared to a more complex model with one additional predictor (Model 1-4) and finally this more complex model is compared to the full Transformed Model. The fifth *F*-test consists only of a single model comparison between the Basic Model and the Transformed Model.

From Table 7 it can be seen that the *p*-value of most of the *F*-tests are above the significance level of 0.05. However, the *p*-value is below the significance level of 0.05 when comparing Model 2-4 with the transformed model. This indicates that at least one of the slope estimates for the additional predictors might not be equal to 0. But when each of Model 1-4 are being compared to the basic model, the *p*-value is above the significance level, which indicates that the null-hypothesis can not be rejected and that the additional predictors are redundant. This is also the case when the transformed model is compared directly to the basic model. And due to parsimony of model fit, it is preferable to choose a simpler model, which the Basic Model is, over a more complex one if possible. By those reasons the Basic Model will be considered for the rest of the paper. A summary of the Basic Model appears from Table 8.

	Estimate	<i>p</i>-value
Intercept	-0.5280	0.0432
$\sqrt{\text{Pairs}}$	1.0159	$5.92 \cdot 10^{-11}$
Size	-0.4955	0.0014
Status	0.4264	0.0096
Residual standard error	0.5647	-
Adjusted R^2	0.6103	-
F -statistic	31.8	$3.913 \cdot 10^{-12}$

Table 8: Properties of the Basic Model

3 Regression diagnostics - Anna

After fitting the model, it is important to once more test it against the assumptions of the linear regression. They help to confirm the validity of the model and allow for a conclusion that the model is reliable in making predictions and drawing conclusions.

3.1 Normality

The first assumption, normality, can be checked using a qq-plot of the residuals (see Figure 4). The plot suggests that the standardised residuals approximately follow a normal distribution. It has thin tails and a slight deviation from what we were expecting in the large values of the residuals. This might be caused by a relatively small sample - 62 observations before removing outliers. It suggests that the model is worse at accurately predicting values when the extinction time is very high. From Figure 4 it is also visible that the final, fitted model follows the assumption of normality much more closely than the full model, even after removing the outliers/influential observations.

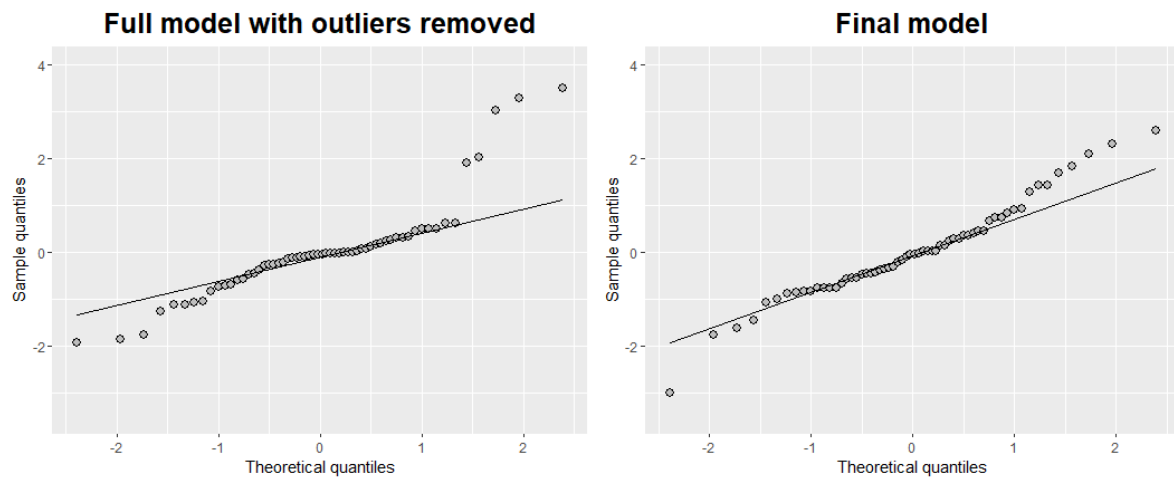


Figure 4: Comparison of standardised residuals between full and fitted models

3.2 Homoscedasticity and linearity

The spread-level (see Figure 5) plot allows for an inspection of whether the fitted model follows the assumption of homoscedasticity and linearity.

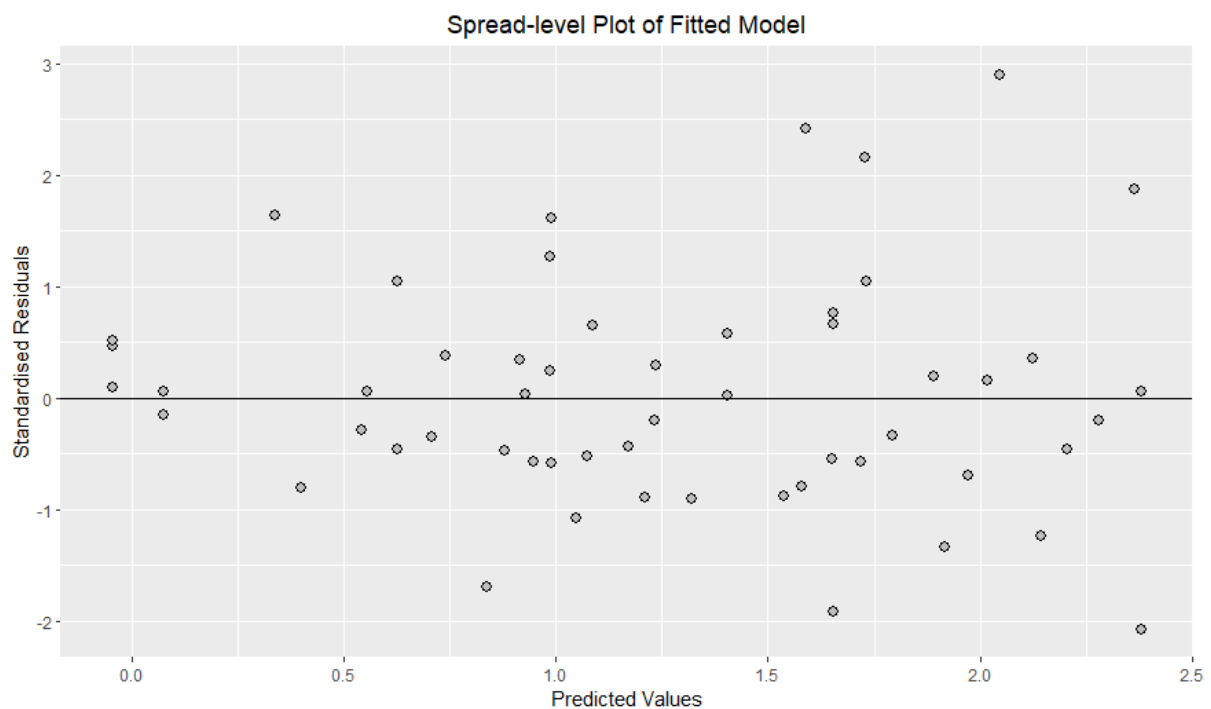


Figure 5: Spread-level plot of the residuals of the fitted model

As we can see, the standardised residuals are in general spread evenly throughout the graph. There is a slight variance between small and large values, suggesting slight heteroscedasticity. This again might be a result of a small sample. The level of heteroscedasticity is not that significant, therefore, we can say that the assumption of homoscedasticity is met. The standardised residuals also do not seem to follow any pattern, being spread randomly, so the assumption of linearity is met as well.

3.3 Outliers and influential observations

Section 2.2 analyzed the outliers and argued for their removal. Figure 6 shows the spread level plot of the full model compared to the final, fitted model. Some outliers are still present in the model, however, given the small number of total observations, and the other reasons explained in Section 2.2, the decision was made not to remove more outliers from the data set.

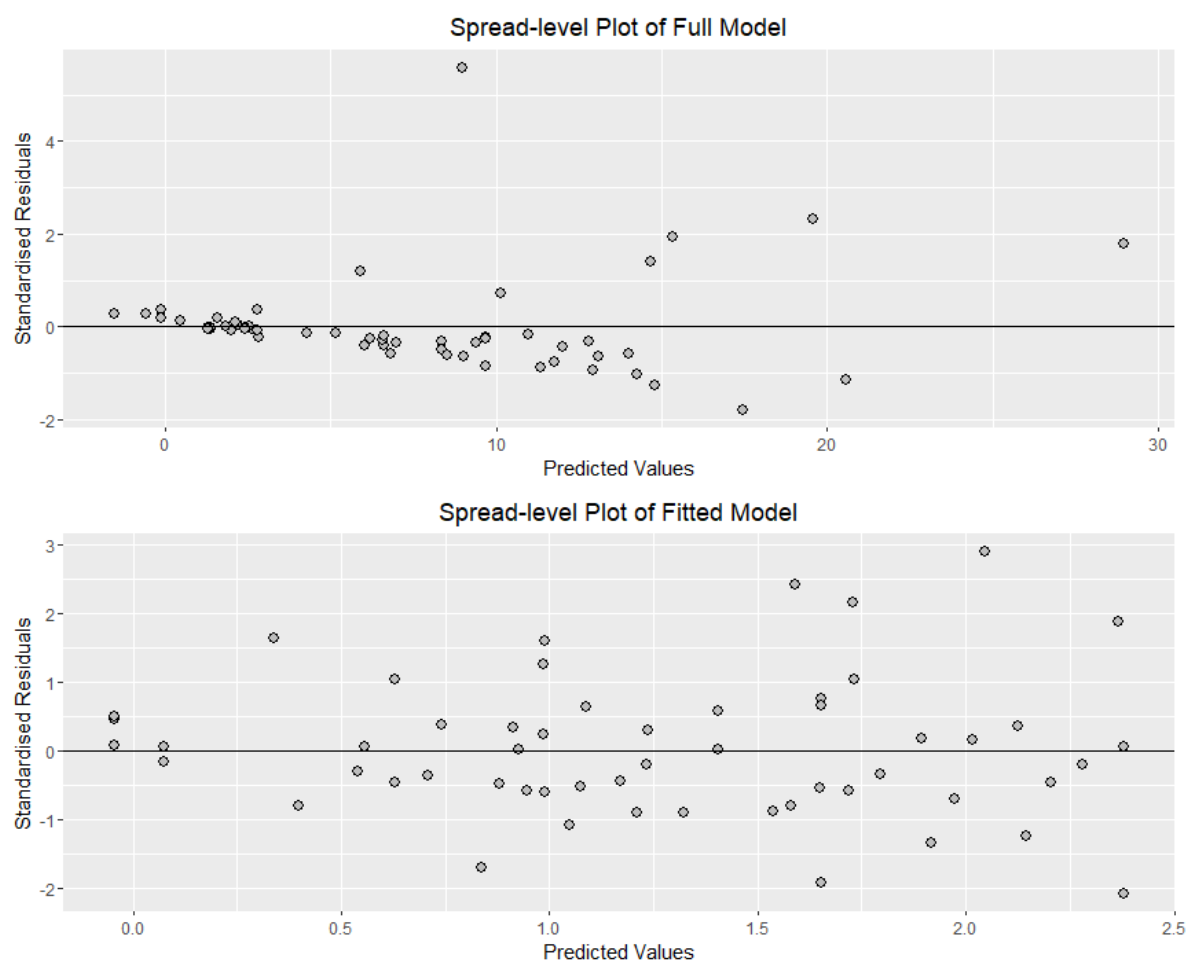


Figure 6: Models comparison of standardised residuals

3.4 Multicollinearity

Collinearity or multicollinearity usually occurs when the predictor variables are so connected with each other that one can be almost perfectly predicted by the other. The predictors in this study seem to be very different, however, it is a good practice to formally check that collinearity is not an issue, especially in the multivariant regression model. Table 9 shows the VIF analysis of the fitted model. The VIF values for all the predictor variables are well below 10 (a number that would indicate too much collinearity), so there is no concern about it in the model.

Predictor variable	VIF-value
Pairs	1.029802
Size	1.001193
Status	1.029006

Table 9: VIF-values of the predictor variables in the fitted model

4 Inference - Sebastian

After having removed influential observations, considered various transformations of the variables and compared the model with a series of nested models, the following model is achieved:

$$\log(\text{Time}) = -0.528 + 1.0159 \cdot \sqrt{\text{Pairs}} - 0.4955 \cdot \text{Size} + 0.4264 \cdot \text{Status}$$

The transformations of the outcome and the Pairs variable make the interpretation of the model a little more complicated. But if we take a look at the slope estimate of the square root transformed Pairs variable it is estimated to be 1.0159. This can be interpreted as a 1-unit change in $\sqrt{\text{Pairs}}$ leads to a 1.0159 change in $\log(\text{Ave. ext. time})$. Since the logarithm in this case has base e , it is a fairly large change, as a 1-unit change in the log-transformed variable corresponds to multiplying by a factor of $e \approx 2.718$. However it requires a big increase in Pairs to obtain a 1-unit change in $\sqrt{\text{Pairs}}$, especially for large numbers of Pairs, since the difference between $\sqrt{a}$ and $\sqrt{b}$ is larger for small values of a and b than for larger values. By relating this relationship to the actual variables, a change in number of nesting pairs has more impact if the number of nesting pairs was initially low. For instance a change from 1 to 11 nesting pairs will result in a larger increase in extinction time than an increase from 100 to 110 nesting pairs.

The slope estimate of the Size predictor is estimated as -0.4955. The Size variable is a dummy variable and the small birds, denoted by S, have been set as the value 1, while the large birds, denoted by L, corresponds to the value 0. Thus the interpretation is that small birds have a lower average extinction time than the larger birds corresponding to a decrease of approximately $e^{\frac{1}{2}}$ years, because of the logarithmic transformation of the outcome. This result seems reasonable, as small birds might be threatened by more species than larger birds and might be more vulnerable to harsh weather conditions.

Another dummy variable is the Status predictor. In this case it is the resident birds (R), which are

considered as the value 1, while migrant birds (M) are denoted by the value 0. Since the slope estimate of the Status variable is 0.4264 the resident birds tend to have higher extinction time by a factor of a little less than $e^{\frac{1}{2}}$ years. Again this seems like a realistic interpretation compared to real life, since migrant birds are exposed to more dangers, when migrating from one location to another.

Overall the effects of the intercept, Size - and Status variables are of less impact, as they do not scale. The intercept is a constant, while it is only possible to gain the effects of the Size and Status slope estimates once. In contradiction, it is possible to scale the effect of the $\sqrt{\text{Pairs}}$ variable. In theory it is always possible to increase the number of nesting pairs, but as mentioned, the effect of a positive change in number of nesting pairs decreases as the actual number of nesting pairs increases. This is due to the slope of the square root function, $\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$, which converges towards 0 as x goes towards ∞ . So a fair assessment from the model would be that the Pairs variable is most crucial for the average extinction time of birds. This statement is supported by the p -value of the $\sqrt{\text{Pairs}}$ variable, which is equal to $5.92 \cdot 10^{-11}$ (see Table 8), which indicates that the slope estimate of this variable is very likely to be equal to the true slope estimate.

5 Conclusion - Snow

To summarize, the final model suggests that the extinction time of birds is explained by the number of nesting pairs, their size and migratory status. Among these predictors, the number of nesting pairs is the most important predictor, as it has the absolute largest slope estimate. Additionally it is possible to scale the number of nesting pairs in contradiction to the size and migratory status, which are dummy variables.

The more nesting pairs a bird species has, the larger the extinction time of that particular species tend to be. However the impact decreases as the number of nesting pairs increases. When it comes to the dummy variables, large residential birds have a longer extinction time compared to small migrating birds, according to the model.

As mentioned in the introduction, the data, which the model fits and tries to explain, is collected among birds on 16 British islands. This sample is not representative for the entire bird population, why the model and this paper does not give a final explanation of which factors are affecting bird extinction. Maybe the model does not apply for birds in other geographically regions or possibly other variables have to be considered. These options remains as unanswered questions and could be investigated by doing further research.